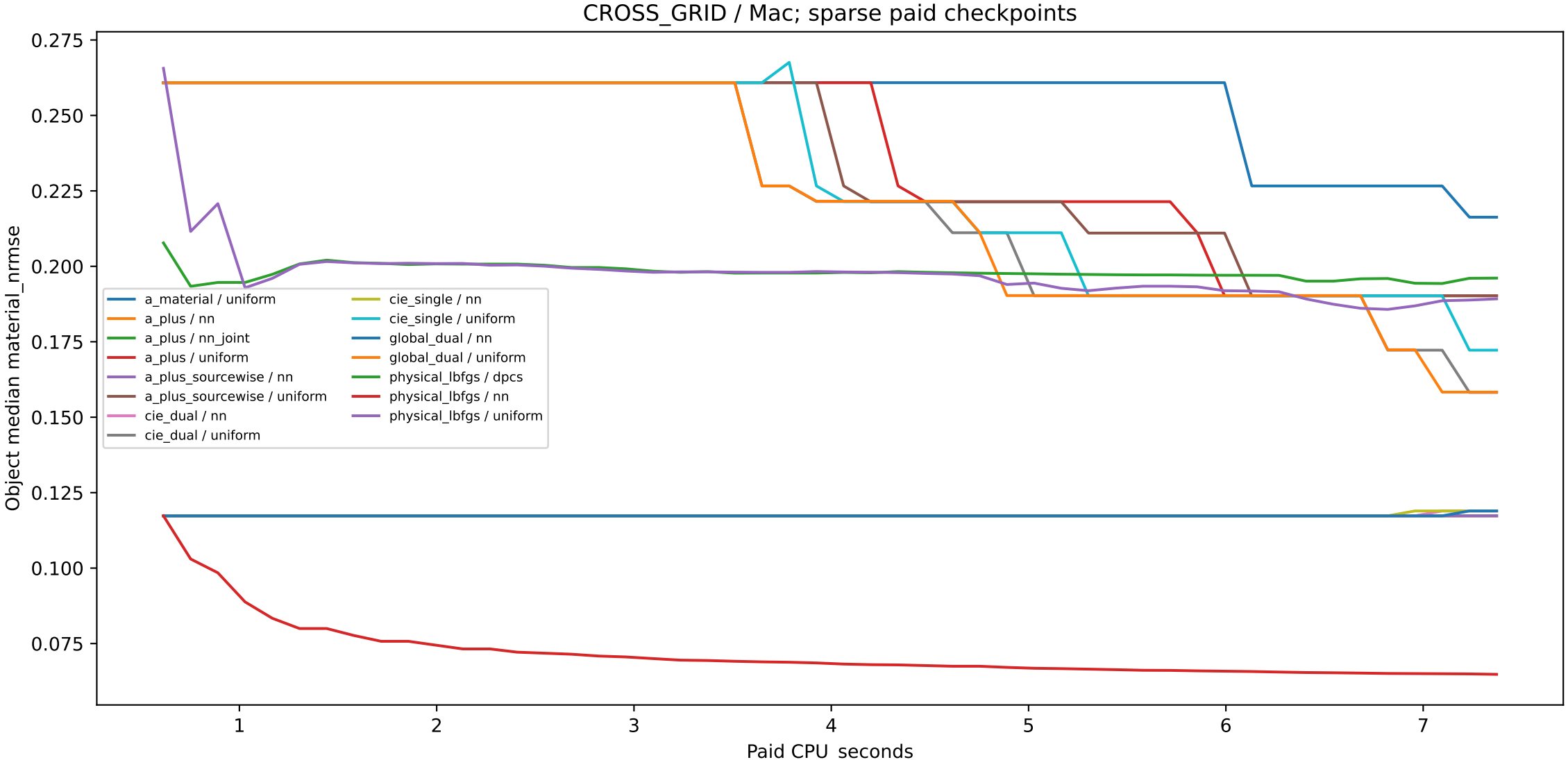
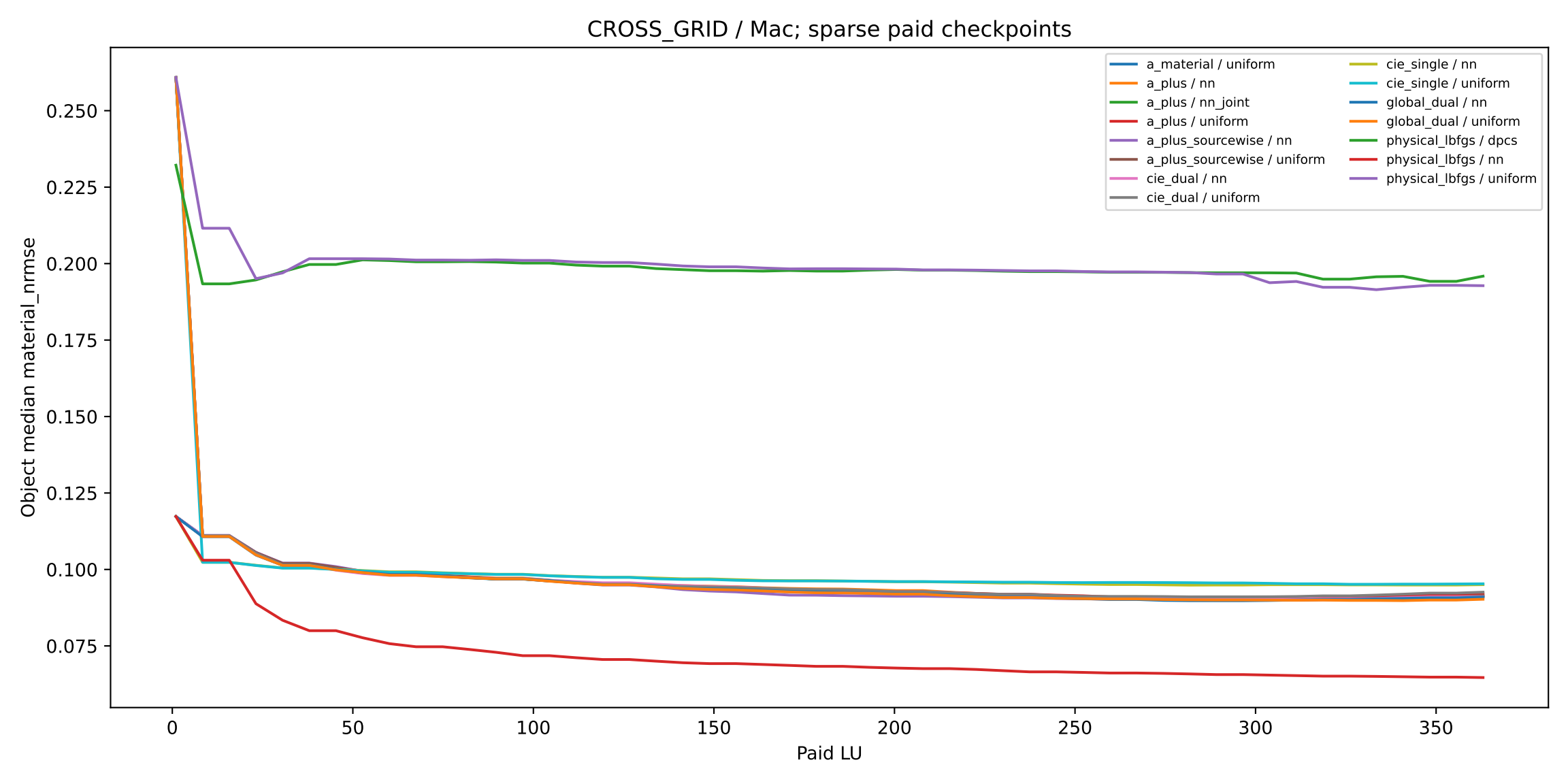
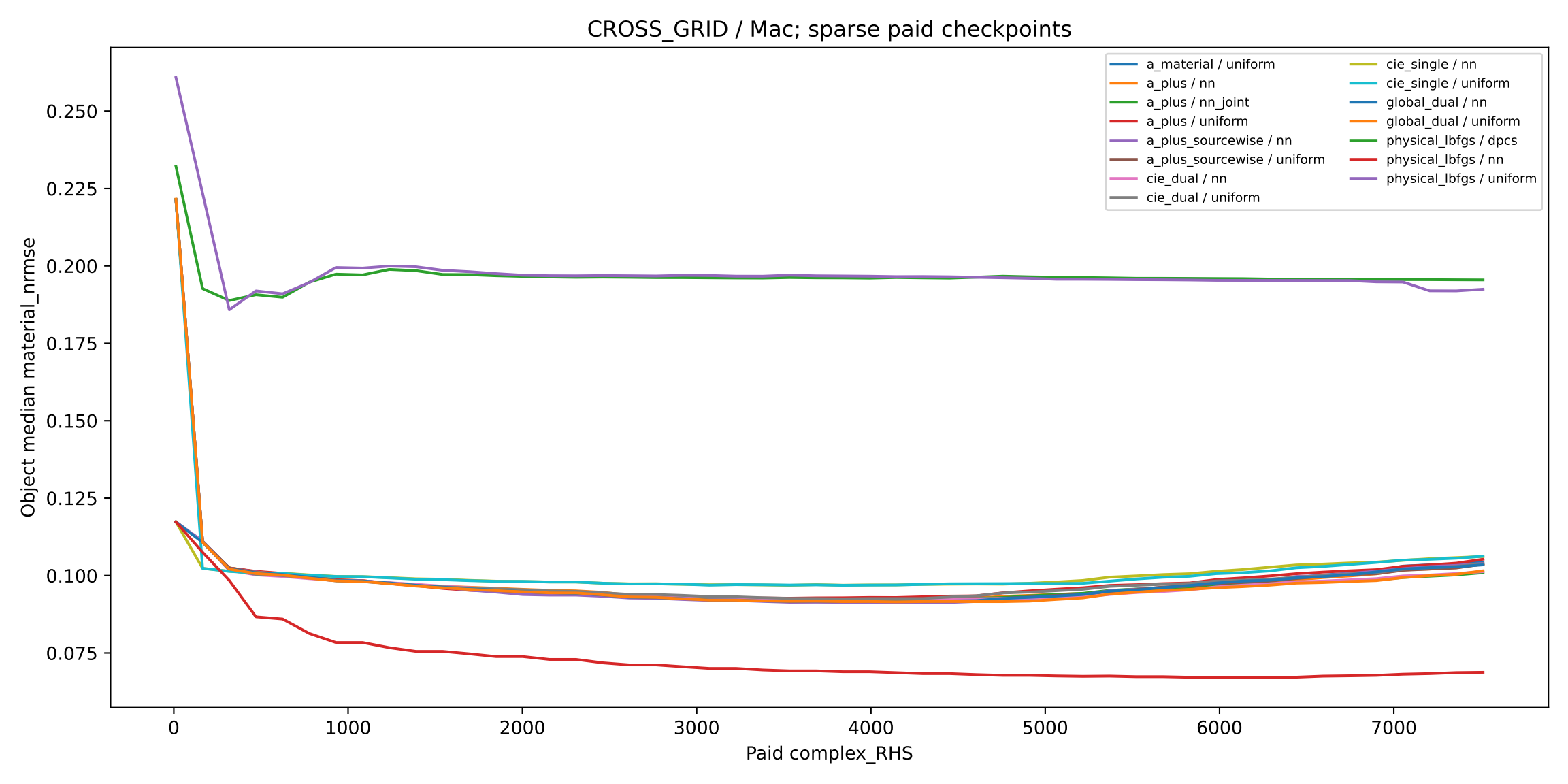
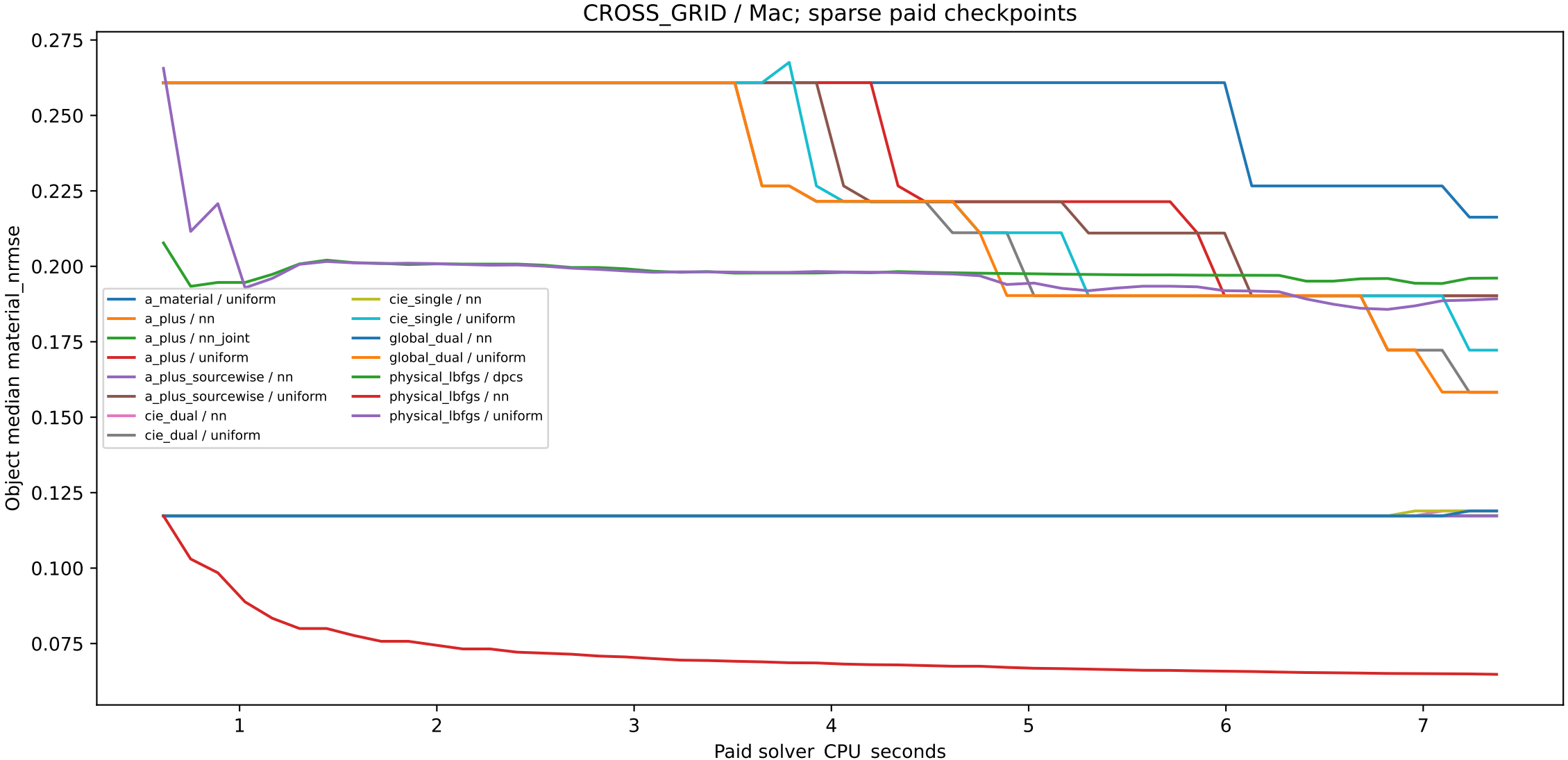
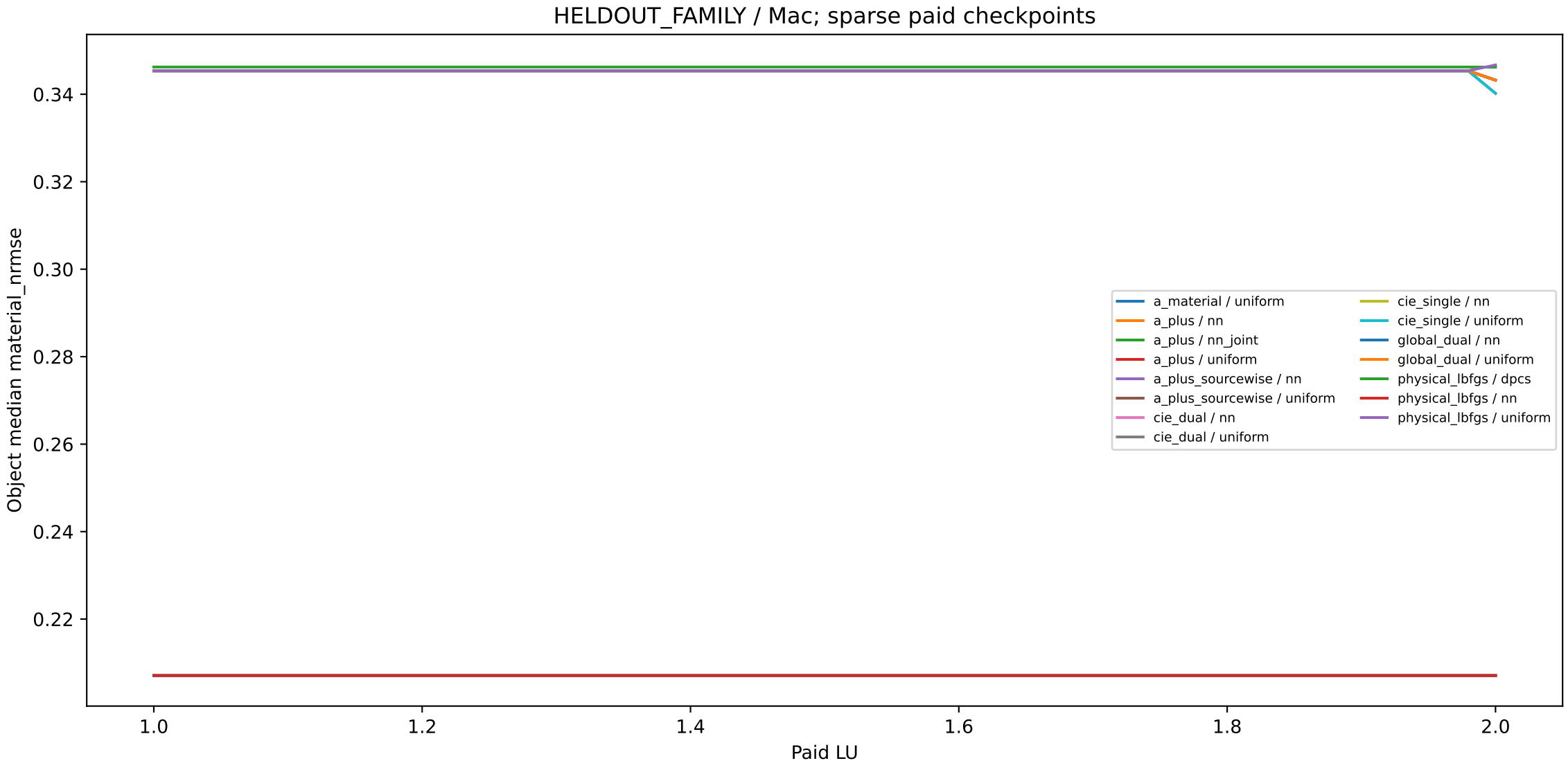
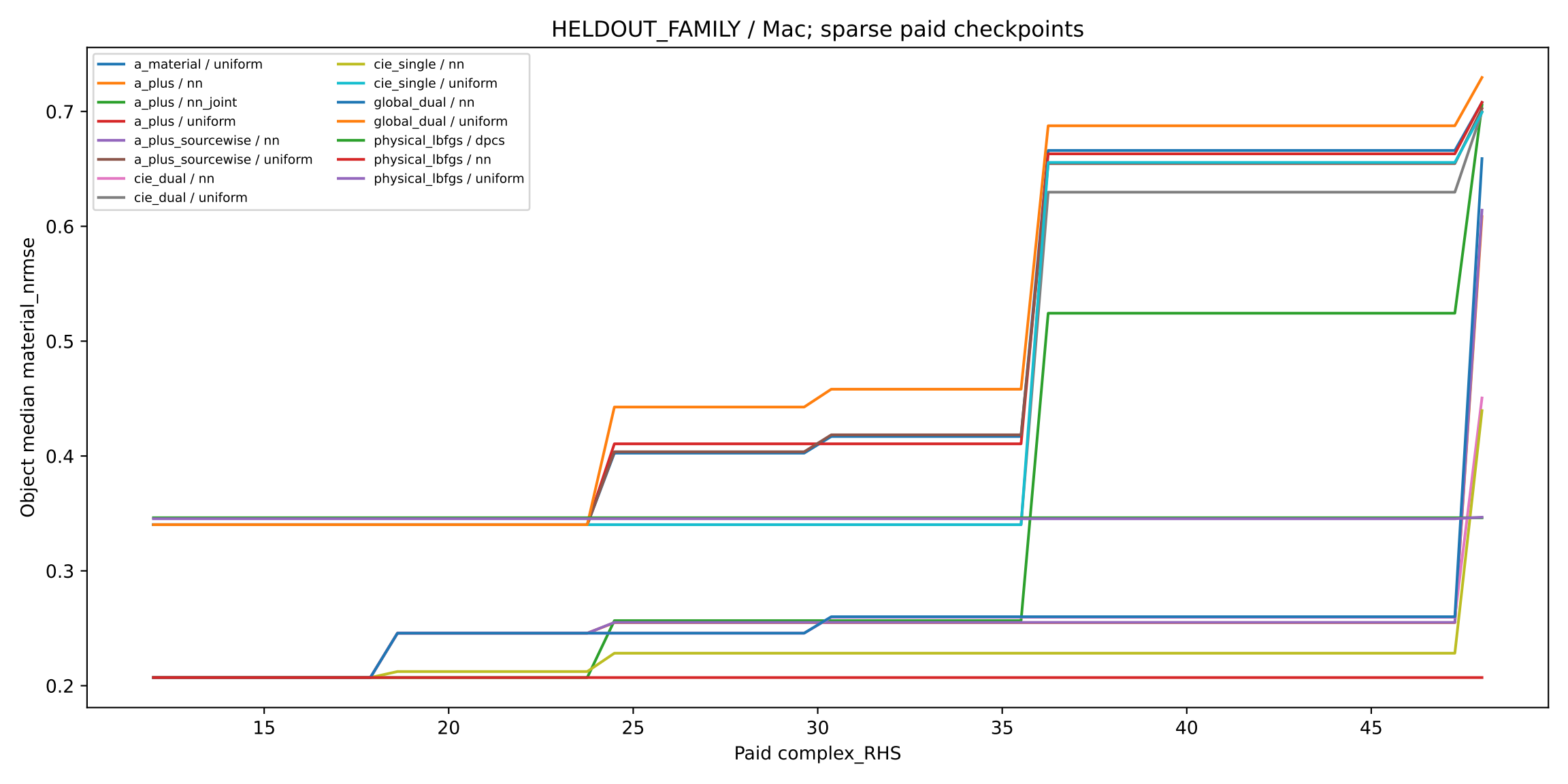

CROSS_GRID / Mac; sparse paid checkpoints
Object median material_nrmse
Paid CPU_seconds
0.275
0.250
0.225
0.200
0.175
0.150
0.125
0.100
0.075
1
2
3
4
5
6
7
a_material / uniform
a_plus / nn
a_plus / nn_joint
a_plus / uniform
a_plus_sourcewise / nn
a_plus_sourcewise / uniform
cie_dual / nn
cie_dual / uniform
cie_single / nn
cie_single / uniform
global_dual / nn
global_dual / uniform
physical_lbfgs / dpcs
physical_lbfgs / nn
physical_lbfgs / uniform

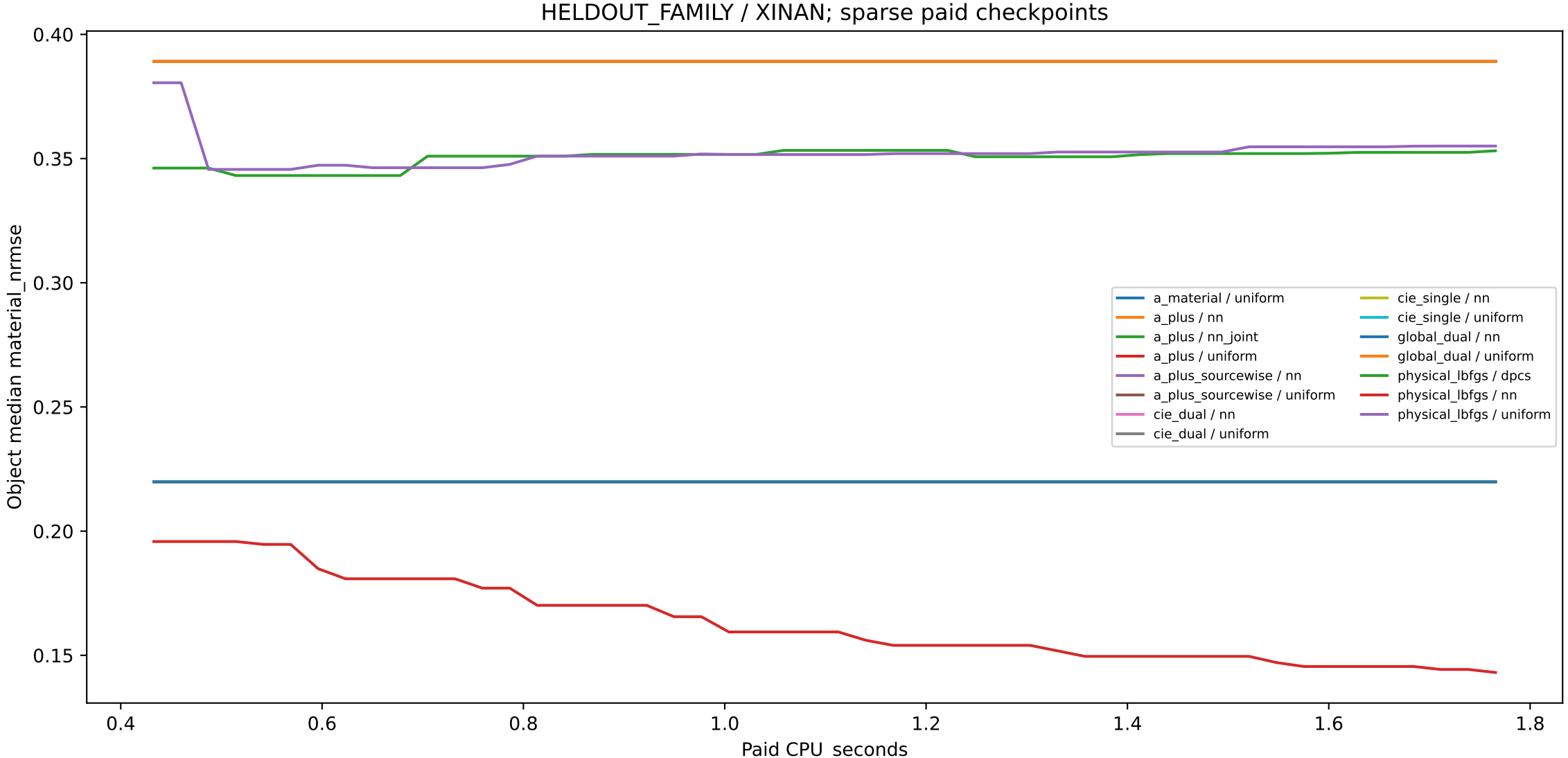

CROSS_GRID / Mac; sparse paid checkpoints
Object median material_nrmse
Paid LU
0.250
0.225
0.200
0.175
0.150
0.125
0.100
0.075
0
50
100
150
200
250
300
350
a_material / uniform
a_plus / nn
a_plus / nn_joint
a_plus / uniform
a_plus_sourcewise / nn
a_plus_sourcewise / uniform
cie_dual / nn
cie_dual / uniform
cie_single / nn
cie_single / uniform
global_dual / nn
global_dual / uniform
physical_lbfgs / dpcs
physical_lbfgs / nn
physical_lbfgs / uniform

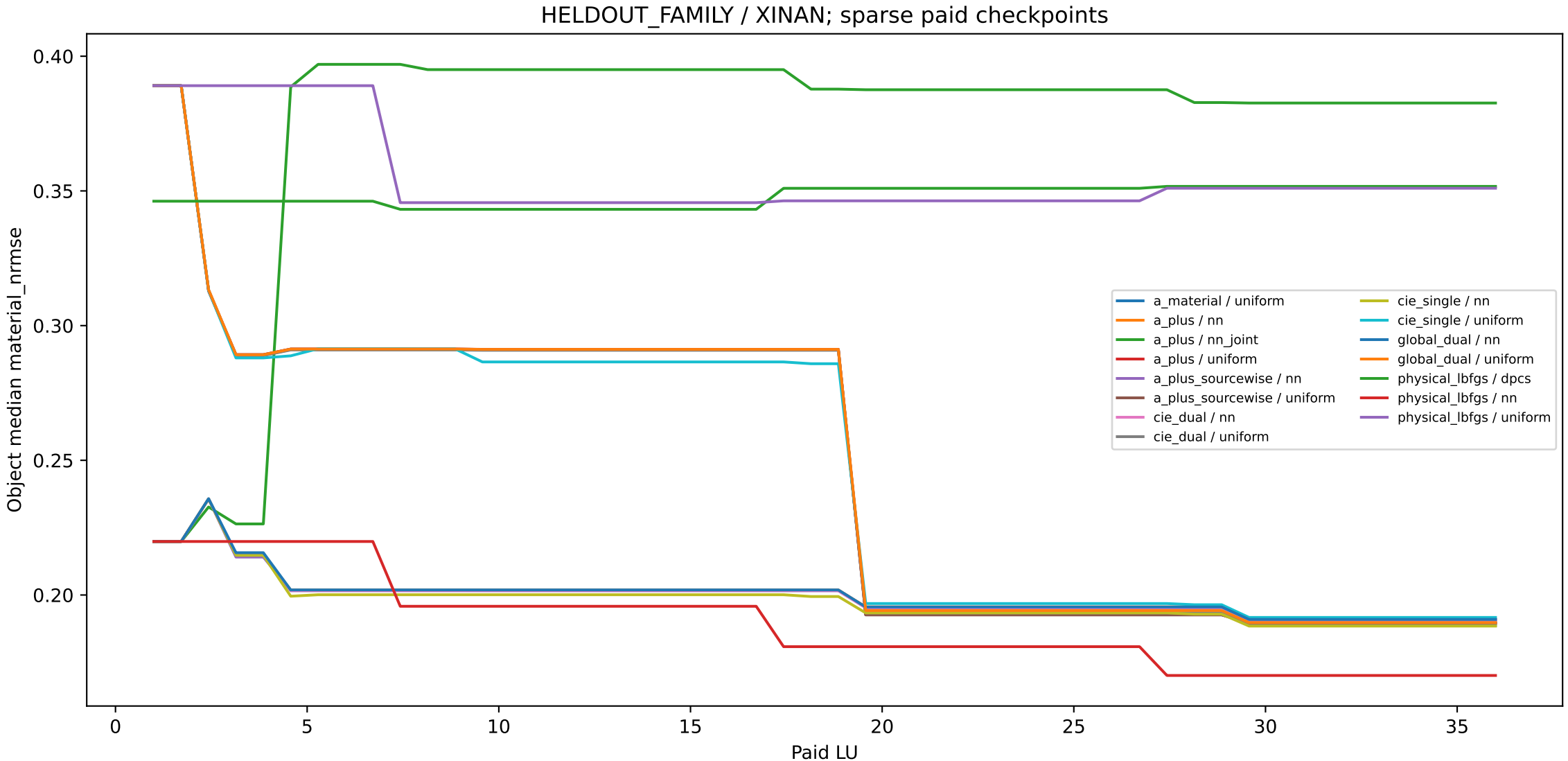

CROSS_GRID / Mac; sparse paid checkpoints
Object median material_nrmse
Paid complex_RHS
0.250
0.225
0.200
0.175
0.150
0.125
0.100
0.075
0
1000
2000
3000
4000
5000
6000
7000
a_material / uniform
a_plus / nn
a_plus / nn_joint
a_plus / uniform
a_plus_sourcewise / nn
a_plus_sourcewise / uniform
cie_dual / nn
cie_dual / uniform
cie_single / nn
cie_single / uniform
global_dual / nn
global_dual / uniform
physical_lbfgs / dpcs
physical_lbfgs / nn
physical_lbfgs / uniform

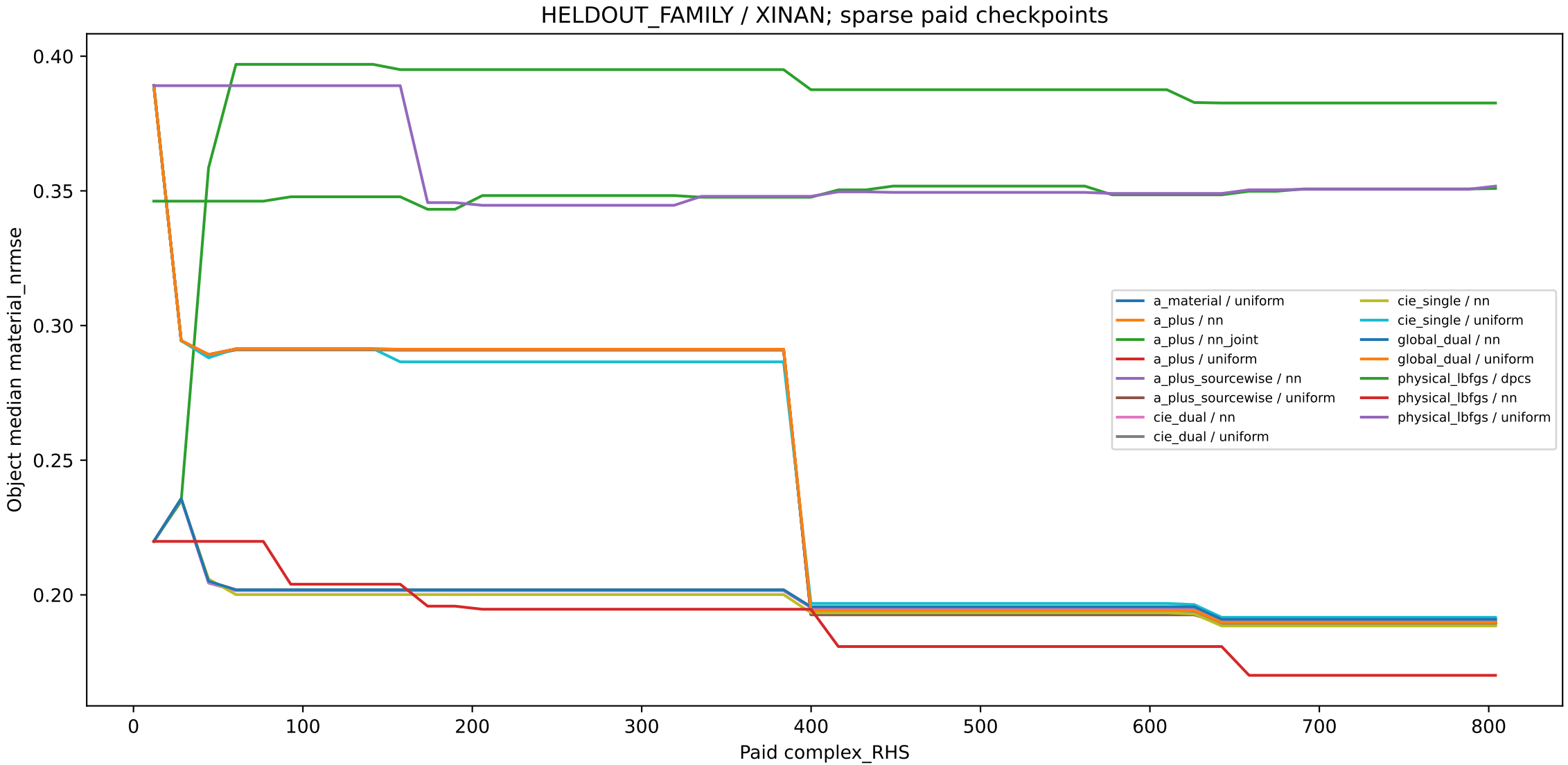

CROSS_GRID / Mac; sparse paid checkpoints
Object median material_nrmse
Paid solver_CPU_seconds
0.275
0.250
0.225
0.200
0.175
0.150
0.125
0.100
0.075
1
2
3
4
5
6
7
a_material / uniform
a_plus / nn
a_plus / nn_joint
a_plus / uniform
a_plus_sourcewise / nn
a_plus_sourcewise / uniform
cie_dual / nn
cie_dual / uniform
cie_single / nn
cie_single / uniform
global_dual / nn
global_dual / uniform
physical_lbfgs / dpcs
physical_lbfgs / nn
physical_lbfgs / uniform

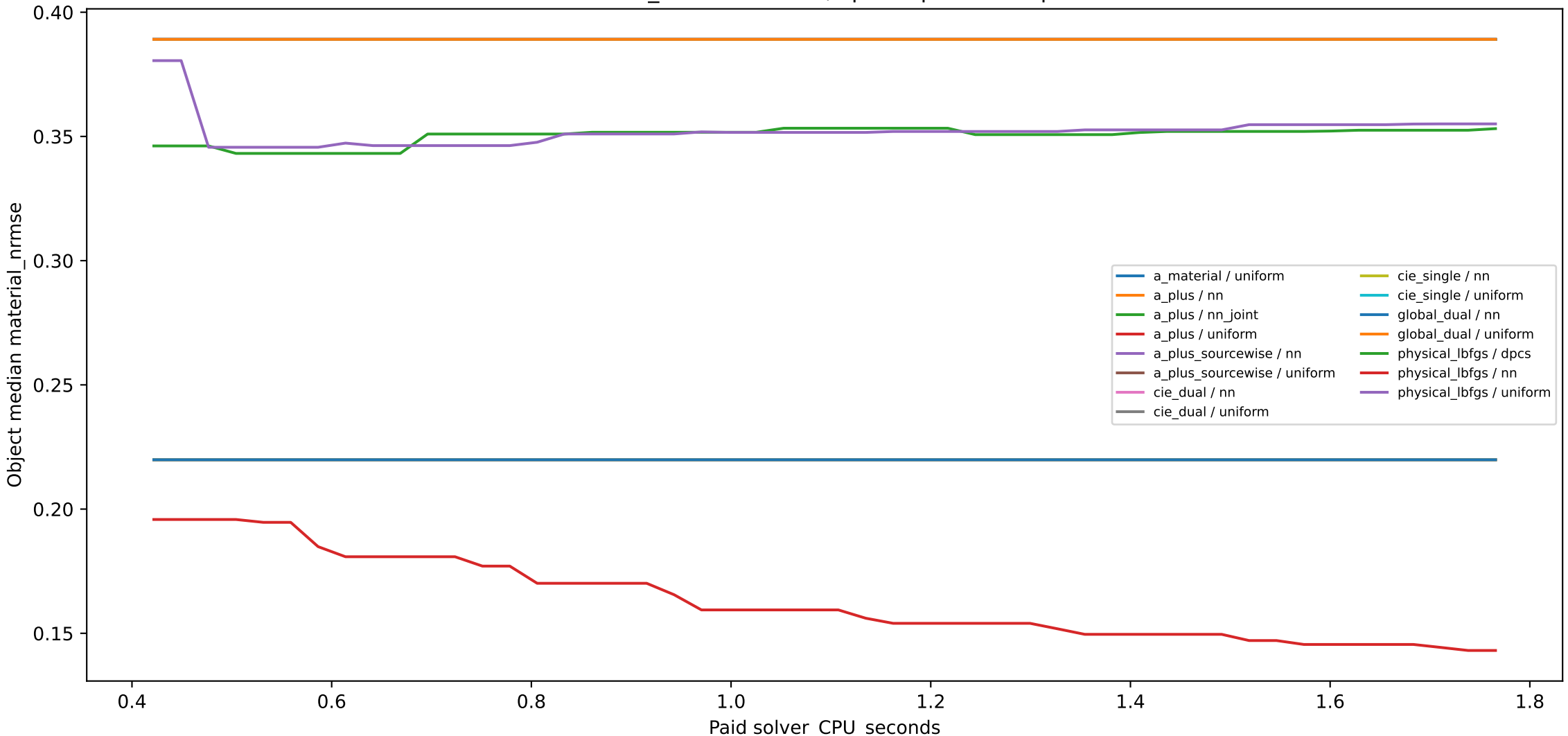

HELDOUT_FAMILY / Mac; sparse paid checkpoints
Object median material_nrmse
0.34
0.32
0.30
0.28
0.26
0.24
0.22
1.0
1.2
1.4
1.6
1.8
2.0
Paid LU
a_material / uniform
a_plus / nn
a_plus / nn_joint
a_plus / uniform
a_plus_sourcewise / nn
a_plus_sourcewise / uniform
cie_dual / nn
cie_dual / uniform
cie_single / nn
cie_single / uniform
global_dual / nn
global_dual / uniform
physical_lbfgs / dpcs
physical_lbfgs / nn
physical_lbfgs / uniform

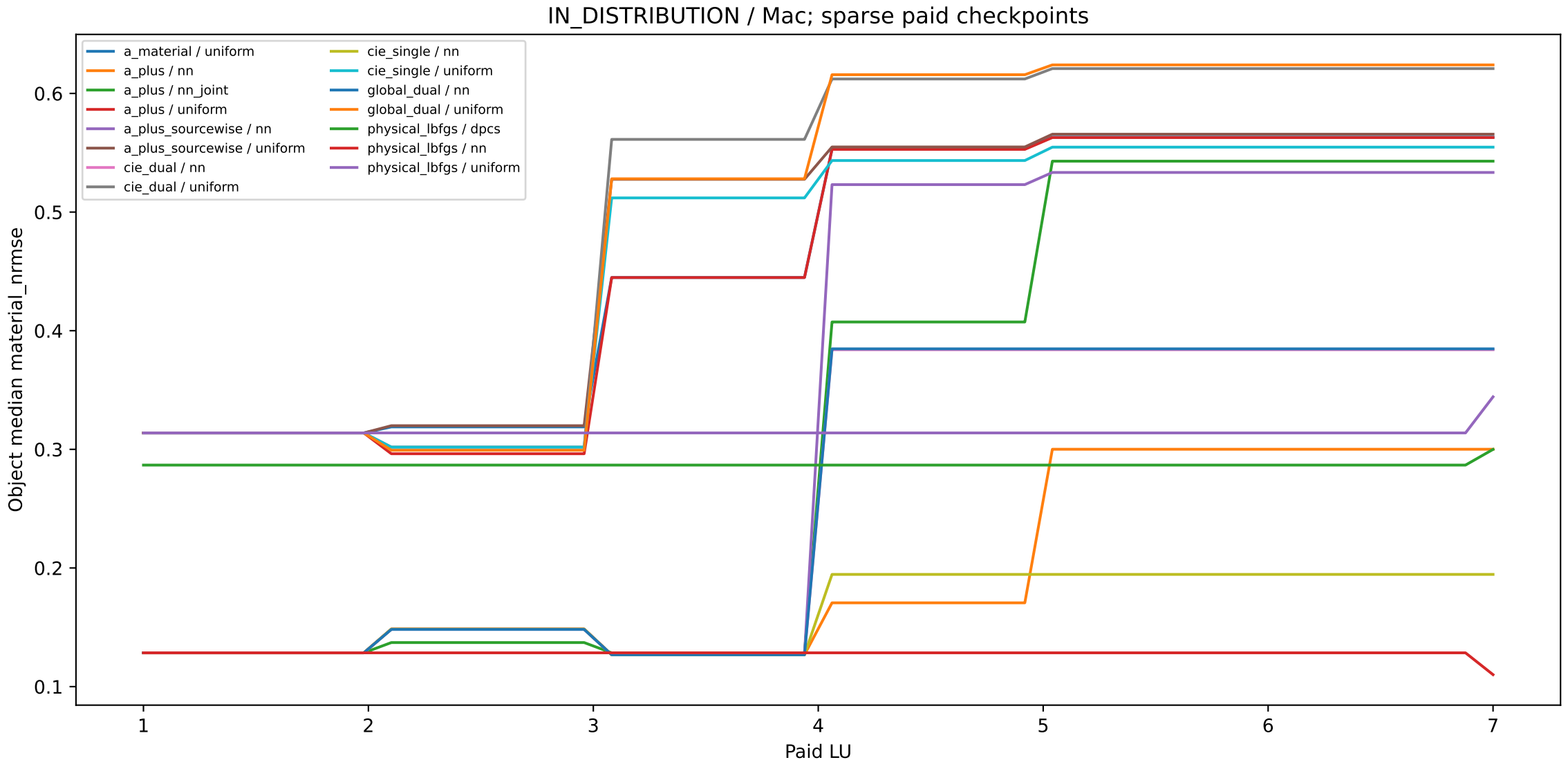

HELDOUT_FAMILY / Mac; sparse paid checkpoints
a_material / uniform
a_plus / nn
a_plus / nn_joint
a_plus / uniform
a_plus_sourcewise / nn
a_plus_sourcewise / uniform
cie_dual / nn
cie_dual / uniform
cie_single / nn
cie_single / uniform
global_dual / nn
global_dual / uniform
physical_lbfgs / dpcs
physical_lbfgs / nn
physical_lbfgs / uniform
Object median material_nrmse
0.2
0.3
0.4
0.5
0.6
0.7
15
20
25
30
35
40
45
Paid complex_RHS

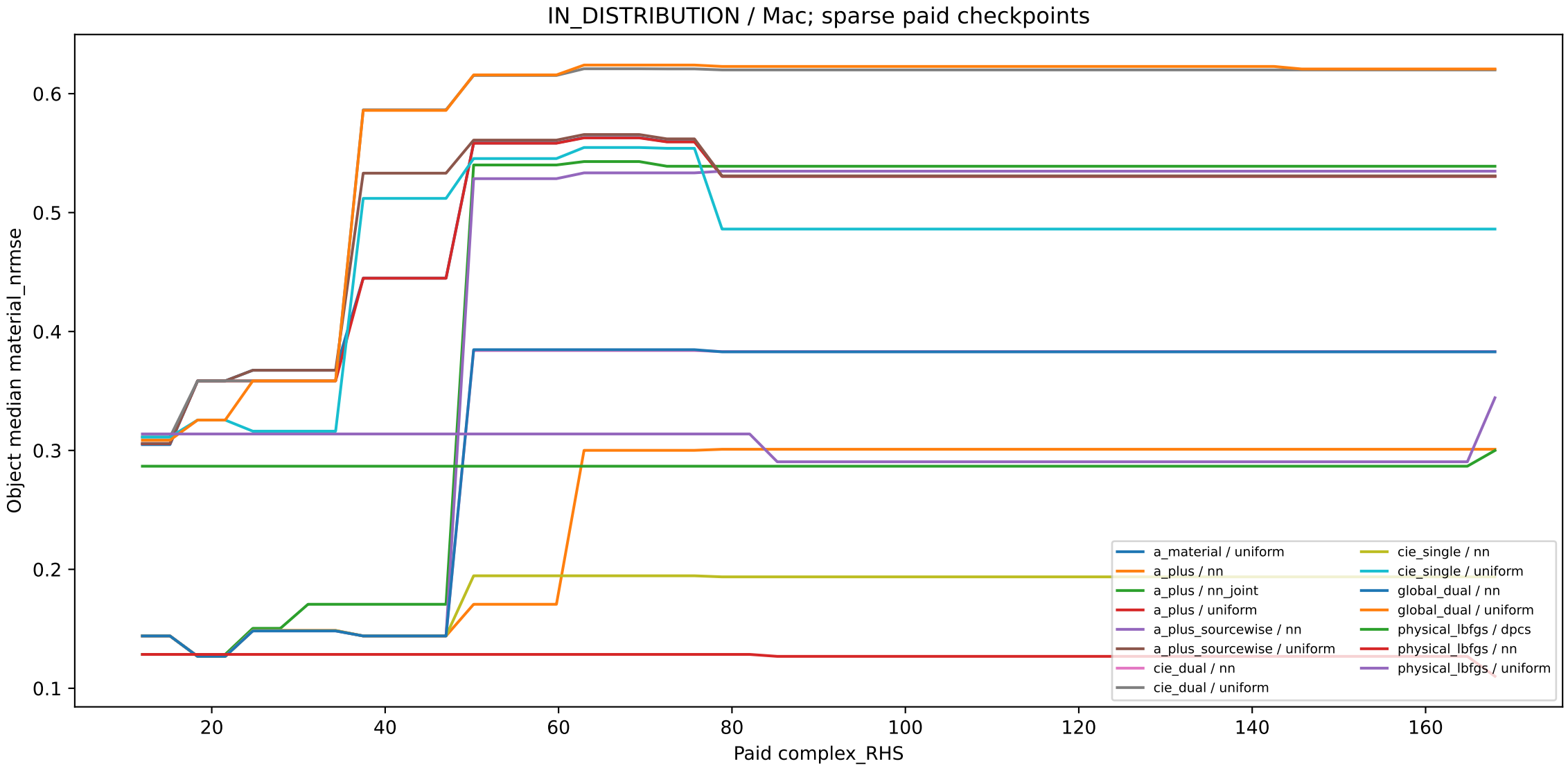

HELDOUT_FAMILY / XINAN; sparse paid checkpoints
Object median material_nrmse
Paid CPU_seconds
0.40
0.35
0.30
0.25
0.20
0.15
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
a_material / uniform
a_plus / nn
a_plus / nn_joint
a_plus / uniform
a_plus_sourcewise / nn
a_plus_sourcewise / uniform
cie_dual / nn
cie_dual / uniform
cie_single / nn
cie_single / uniform
global_dual / nn
global_dual / uniform
physical_lbfgs / dpcs
physical_lbfgs / nn
physical_lbfgs / uniform

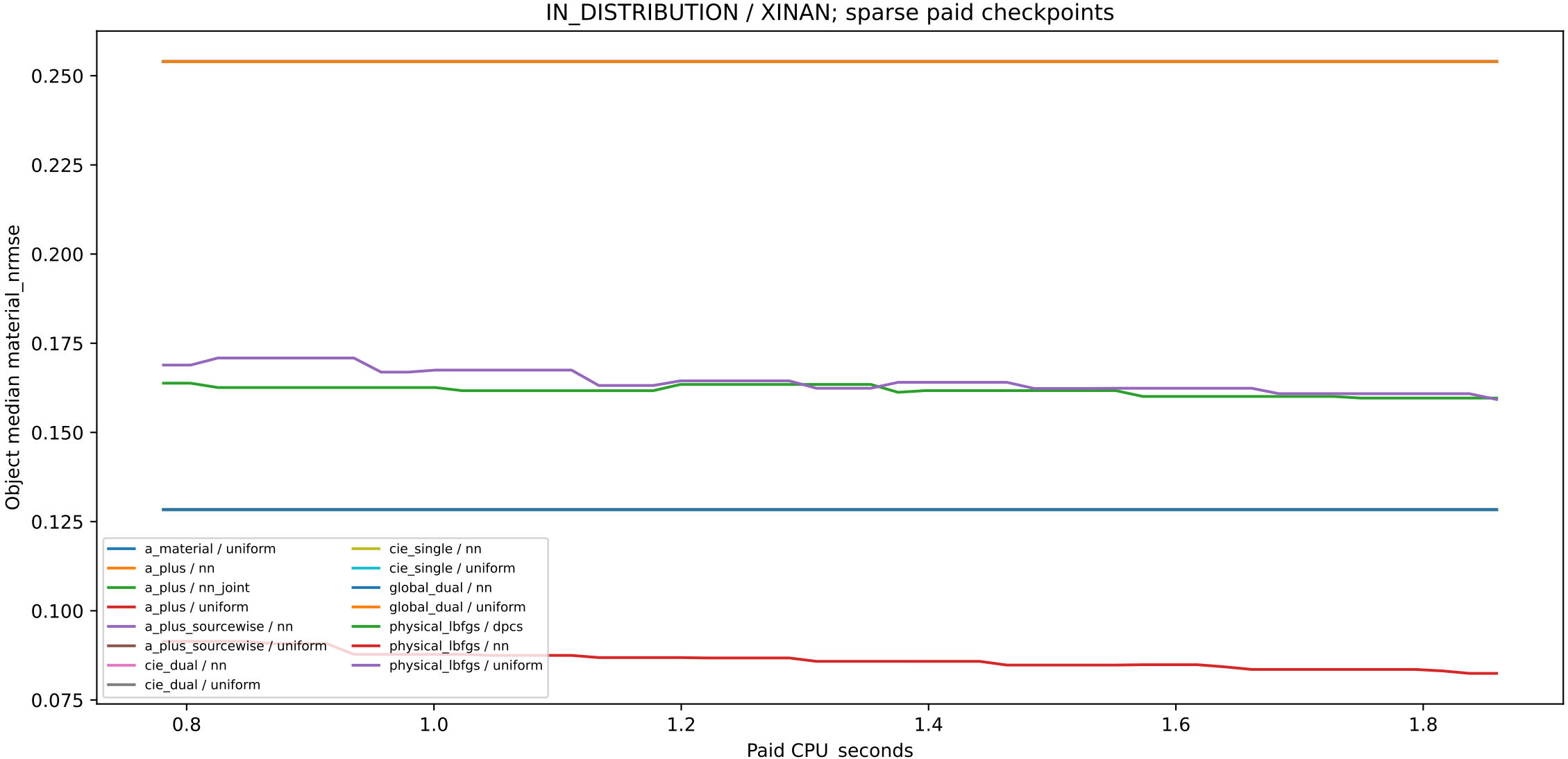

HELDOUT_FAMILY / XINAN; sparse paid checkpoints
Object median material_nrmse
Paid LU
0.40
0.35
0.30
0.25
0.20
0
5
10
15
20
25
30
35
a_material / uniform
a_plus / nn
a_plus / nn_joint
a_plus / uniform
a_plus_sourcewise / nn
a_plus_sourcewise / uniform
cie_dual / nn
cie_dual / uniform
cie_single / nn
cie_single / uniform
global_dual / nn
global_dual / uniform
physical_lbfgs / dpcs
physical_lbfgs / nn
physical_lbfgs / uniform

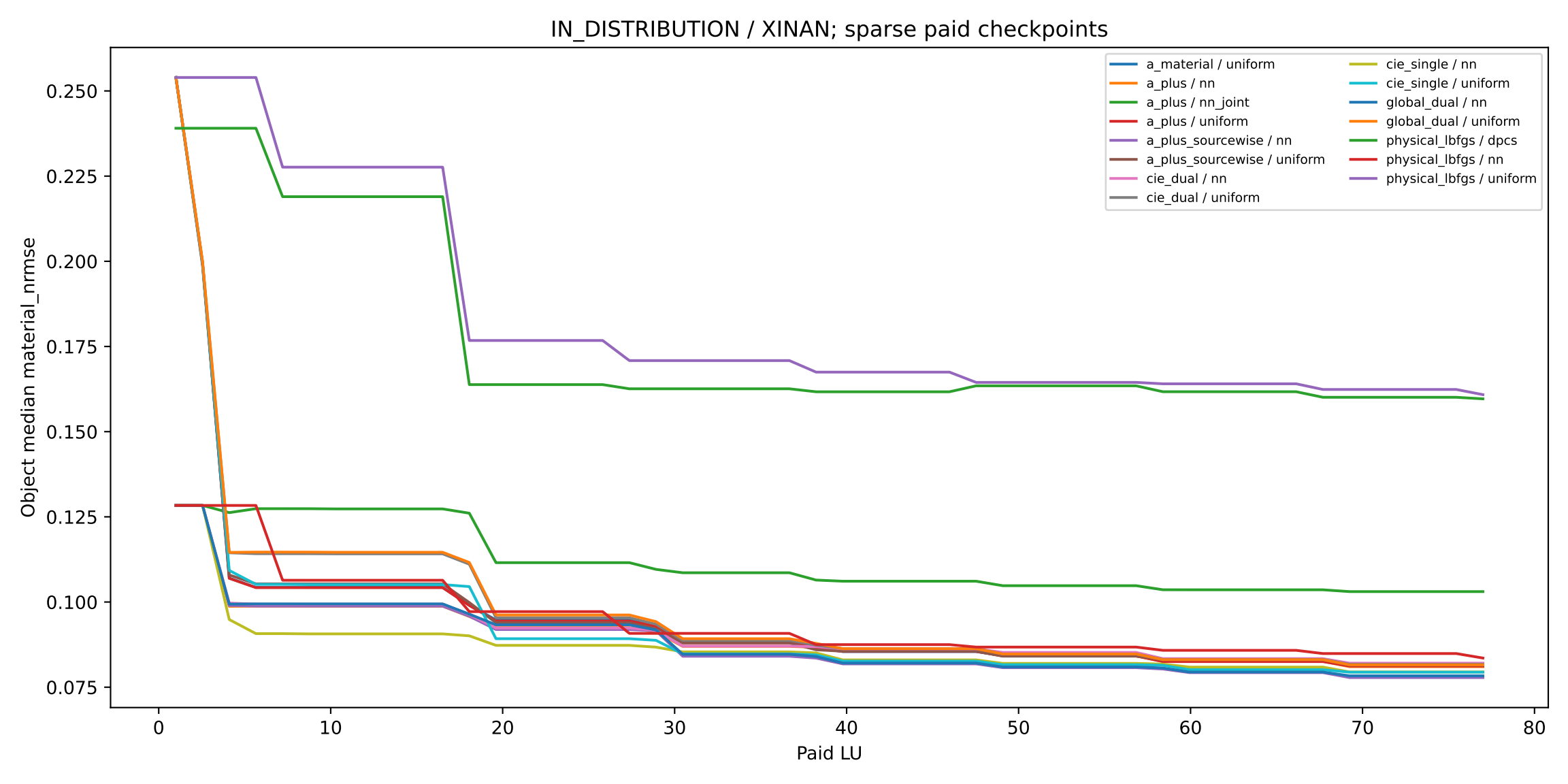

HELDOUT_FAMILY / XINAN; sparse paid checkpoints
Object median material_nrmse
Paid complex_RHS
0.40
0.35
0.30
0.25
0.20
0
100
200
300
400
500
600
700
800
a_material / uniform
a_plus / nn
a_plus / nn_joint
a_plus / uniform
a_plus_sourcewise / nn
a_plus_sourcewise / uniform
cie_dual / nn
cie_dual / uniform
cie_single / nn
cie_single / uniform
global_dual / nn
global_dual / uniform
physical_lbfgs / dpcs
physical_lbfgs / nn
physical_lbfgs / uniform

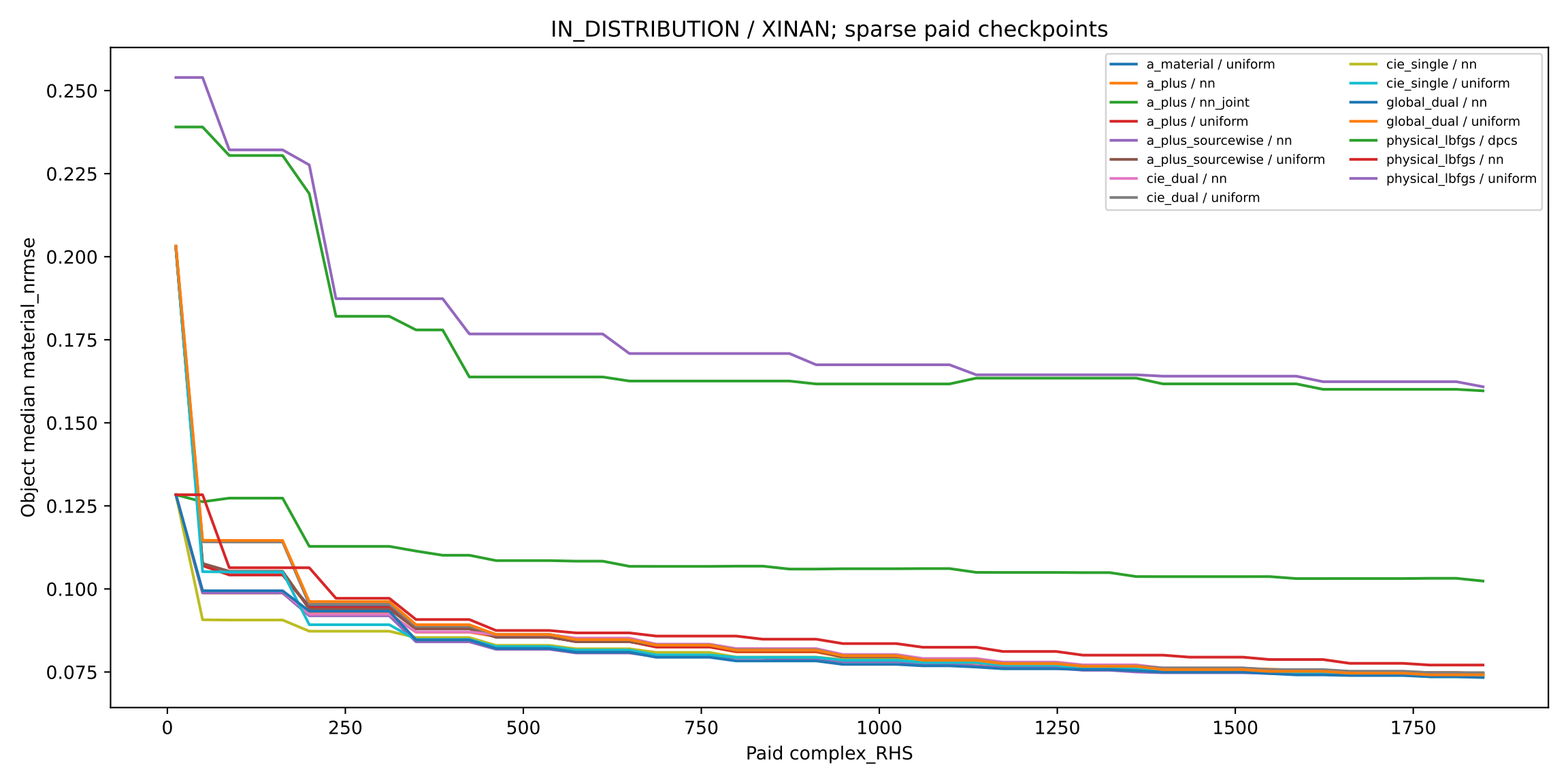

HELDOUT_FAMILY / XINAN; sparse paid checkpoints
Object median material_nrmse
Paid solver_CPU_seconds
0.40
0.35
0.30
0.25
0.20
0.15
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
a_material / uniform
a_plus / nn
a_plus / nn_joint
a_plus / uniform
a_plus_sourcewise / nn
a_plus_sourcewise / uniform
cie_dual / nn
cie_dual / uniform
cie_single / nn
cie_single / uniform
global_dual / nn
global_dual / uniform
physical_lbfgs / dpcs
physical_lbfgs / nn
physical_lbfgs / uniform

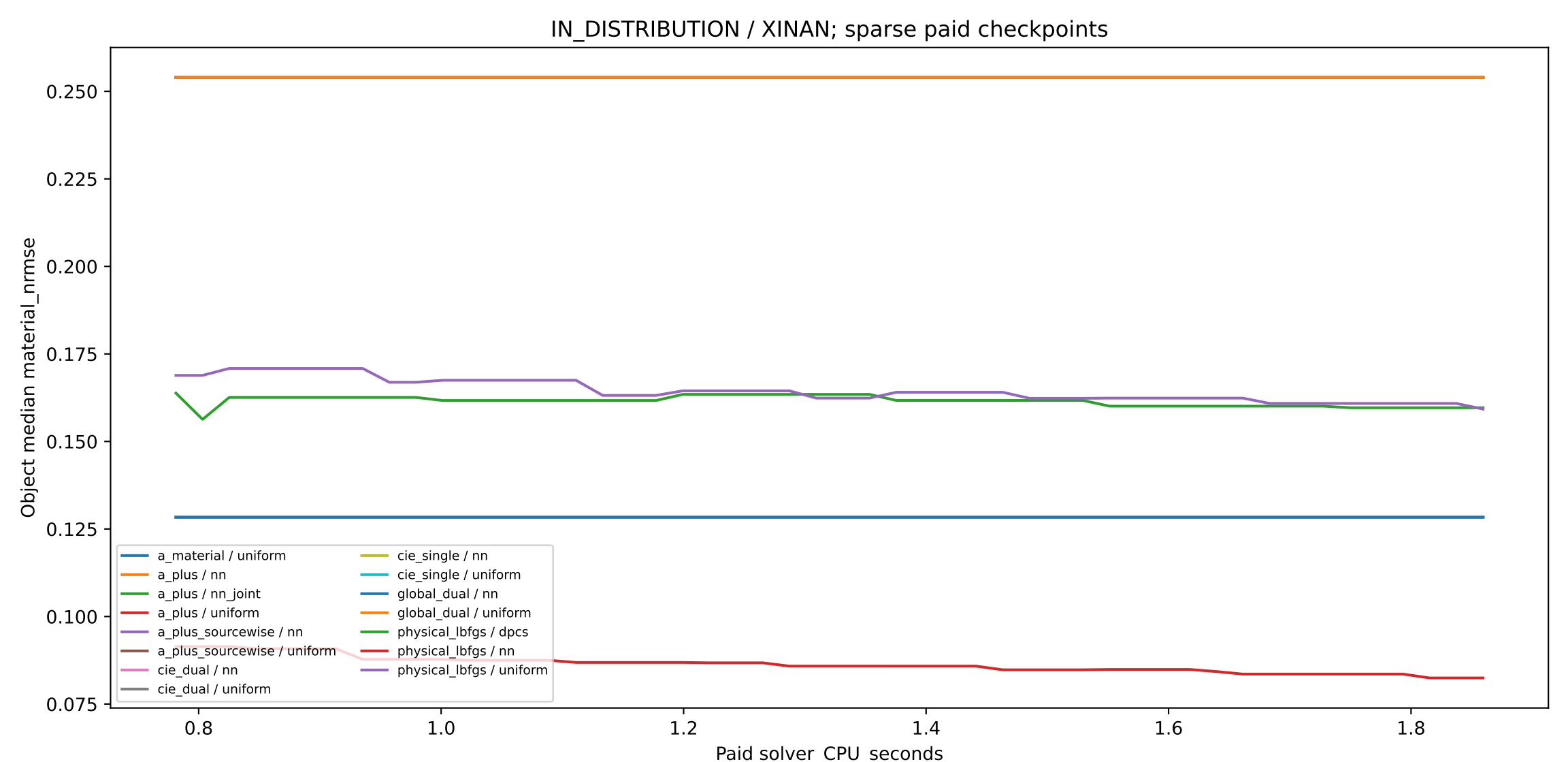

IN_DISTRIBUTION / Mac; sparse paid checkpoints
a_material / uniform
a_plus / nn
a_plus / nn_joint
a_plus / uniform
a_plus_sourcewise / nn
a_plus_sourcewise / uniform
cie_dual / nn
cie_dual / uniform
cie_single / nn
cie_single / uniform
global_dual / nn
global_dual / uniform
physical_lbfgs / dpcs
physical_lbfgs / nn
physical_lbfgs / uniform
Object median material_nrmse
0.1
0.2
0.3
0.4
0.5
0.6
1
2
3
4
5
6
7
Paid LU


IN_DISTRIBUTION / Mac; sparse paid checkpoints
Object median material_nrmse
Paid complex_RHS
0.1
0.2
0.3
0.4
0.5
0.6
20
40
60
80
100
120
140
160
a_material / uniform
a_plus / nn
a_plus / nn_joint
a_plus / uniform
a_plus_sourcewise / nn
a_plus_sourcewise / uniform
cie_dual / nn
cie_dual / uniform
cie_single / nn
cie_single / uniform
global_dual / nn
global_dual / uniform
physical_lbfgs / dpcs
physical_lbfgs / nn
physical_lbfgs / uniform


IN_DISTRIBUTION / XINAN; sparse paid checkpoints
Object median material_nrmse
0.250
0.225
0.200
0.175
0.150
0.125
0.100
0.075
0.8
1.0
1.2
1.4
1.6
1.8
Paid CPU_seconds
a_material / uniform
a_plus / nn
a_plus / nn_joint
a_plus / uniform
a_plus_sourcewise / nn
a_plus_sourcewise / uniform
cie_dual / nn
cie_dual / uniform
cie_single / nn
cie_single / uniform
global_dual / nn
global_dual / uniform
physical_lbfgs / dpcs
physical_lbfgs / nn
physical_lbfgs / uniform


IN_DISTRIBUTION / XINAN; sparse paid checkpoints
Object median material_nrmse
Paid LU
0.250
0.225
0.200
0.175
0.150
0.125
0.100
0.075
0
10
20
30
40
50
60
70
80
a_material / uniform
a_plus / nn
a_plus / nn_joint
a_plus / uniform
a_plus_sourcewise / nn
a_plus_sourcewise / uniform
cie_dual / nn
cie_dual / uniform
cie_single / nn
cie_single / uniform
global_dual / nn
global_dual / uniform
physical_lbfgs / dpcs
physical_lbfgs / nn
physical_lbfgs / uniform


IN_DISTRIBUTION / XINAN; sparse paid checkpoints
Object median material_nrmse
Paid complex_RHS
0.250
0.225
0.200
0.175
0.150
0.125
0.100
0.075
0
250
500
750
1000
1250
1500
1750
a_material / uniform
a_plus / nn
a_plus / nn_joint
a_plus / uniform
a_plus_sourcewise / nn
a_plus_sourcewise / uniform
cie_dual / nn
cie_dual / uniform
cie_single / nn
cie_single / uniform
global_dual / nn
global_dual / uniform
physical_lbfgs / dpcs
physical_lbfgs / nn
physical_lbfgs / uniform


IN_DISTRIBUTION / XINAN; sparse paid checkpoints
Object median material_nrmse
Paid solver_CPU_seconds
0.250
0.225
0.200
0.175
0.150
0.125
0.100
0.075
0.8
1.0
1.2
1.4
1.6
1.8
a_material / uniform
a_plus / nn
a_plus / nn_joint
a_plus / uniform
a_plus_sourcewise / nn
a_plus_sourcewise / uniform
cie_dual / nn
cie_dual / uniform
cie_single / nn
cie_single / uniform
global_dual / nn
global_dual / uniform
physical_lbfgs / dpcs
physical_lbfgs / nn
physical_lbfgs / uniform